EFU SEWAGE & WASTEWATER LICENCE v1.0

Experimental EFU-based Sewage Treatment and Wastewater Framework

0. Purpose and Scope

Experimental EFU-based licensing and measurement framework for sewage treatment (municipal, industrial wastewater, sewer networks, sludge management). **NOT** a water framework directive permit, wastewater treatment operating license, or sludge disposal authorization replacement, but an open research protocol measuring sewage operations' physical footprint in human-scale EFU metrics (energy, chemicals, sludge, nutrient loading). Covers municipal treatment plants, industrial pre-treatment, sewer cleaning.

I. Core Concepts

1. **EFU-Sewage (EFU-SEWAGE)**
   EFU framework application for wastewater treatment:
   - **EFU-E_SEW**: electricity (pumps, mixers, aeration)
   - **EFU-CHEM_SEW**: polymers, chlorine, iron salts (coagulants)
   - **EFU-S_SEW**: sludge volume + dewatering
   - **EFU-N_SEW**: nitrogen, phosphorus (eutrophication risk)
   - **EFU-INFRA_SEW**: sewer maintenance, pipe network amortization
2. **Sewage Sovereignty Gap (SS_SEWAGE)**
   SS_SEWAGE = TEFU_SEWAGE_total – Regional Sewage Carrying Capacity (RSCC_SEWAGE)
   Measures if wastewater treatment stays within regional aquifer + soil assimilation capacity.
3. **MROI_SEWAGE**
   MROI_SEWAGE = THI_SEWAGE / TEFU_SEWAGE
   THI_SEWAGE = public health + water quality + nutrient recovery.

II. EFU Sewage Licence – Research Status

Documents EFU-audited wastewater treatment plants, sewerage operations, sludge processors.
Does NOT replace UWWTD compliance, water quality permits, or sludge quality certificates.
Enables activated sludge vs. membrane bioreactor benchmarking.

III. Metrics and Thresholds

1. EFU Components (per population equivalent - PE)

EFU-E_SEW (Energy)
EFU-E_SEW = energy_MWh / 12.88 / 1000 PE/year
Example: Conventional = 40 kWh/PE/year → 0.00031 EFU/PE

EFU-CHEM_SEW (Chemicals)
EFU-CHEM_SEW = coagulant_kg / 10kg/person/year / 1000 PE

EFU-S_SEW (Sludge)
EFU-S_SEW = sludge_m³ / 100m³/person/year / 1000 PE

TEFU_SEWAGE = 1.3×EFU-E + 1.6×EFU-CHEM + 2.0×EFU-S + 1.8×EFU-N + 1.1×EFU-INFRA

2. Regional Sewage Carrying Capacity (RSCC_SEWAGE)

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RSCC_SEWAGE = Sustainable_PE / Population / 1 PE/person
Example: Germany = 84M PE / 84M = 1.0 EFU_SEWAGE/person/year
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IV. Licence Categories (EFU-SEWAGE Labels)

Level	SS_SEWAGE	MROI_SEWAGE	Label	Interpretation
1	≤ 0	≥1e-2	**EFU-SEWAGE SOVEREIGN**	Natural systems
2	0 < ≤20k EFU	≥5e-3	**EFU-SEWAGE BALANCED**	Biological treatment
3	20k < ≤100k EFU	≥1e-3	**EFU-SEWAGE DEFICIT**	Chemical treatment
4	>100k EFU	<1e-3	**EFU-SEWAGE CRITICAL**	Sludge crisis

V. Regional RSCC_SEWAGE Calibrations

Country	Capacity (M PE)	Population (M)	RSCC_SEWAGE (EFU/person/year)
GERMANY	84	84	1.00
USA	300	340	0.88
CHINA	400	1,410	0.28
JAPAN	125	125	1.00
HUNGARY	9	9.7	0.93

VI. Example: German Wastewater Sector

Input (annual sector):

- Capacity: 84M PE
- Energy: 3.4B MWh (40 kWh/PE)
- Chemicals: 420,000 tonnes
- Sludge: 2.5M tonnes dry matter
- Workers: 50,000 staff
- Region: Germany (RSCC_SEWAGE = 1.0 EFU/person/year)

Calculation:

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EFU-E_SEW = 3.4B/12.88/84M = 0.00031 EFU/PE
EFU-CHEM_SEW = 420k/10/84M = 0.0005 EFU/PE

TEFU_SEWAGE_total ≈ 85,000,000 EFU/year
Allocated = 84M × 1.0 = 84M EFU/year
SS_SEWAGE = 85M – 84M = +1,000,000 EFU/year (Level 3: DEFICIT)

Result: EFU-SEWAGE DEFICIT
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Technology Comparison: Nature-based solutions = 0.15 EFU/PE (Level 2: BALANCED)

VII. Implementation Notes

1. **UWWTD implementation**: TEFU_SEWAGE treatment efficiency metric
2. **P recovery**: EFU-N_SEW fertilizer feedstock prioritization
3. **Sludge biogas**: EFU-E_SEW energy recovery credits
4. **Data sources**: Tier 1 = electricity bills, chemical receipts; Tier 2 = Eurostat water statistics